Wed 26 Mar 2025

Last time we started to discuss an algorithm for finding x^* such that $f(x^*) = 0$ for some given continuous function f.

The algorithm (the Bisection method) depends on repeated application of the intermediate value theorem.

One of the algorithm's features is that we can determine in advance the number of iterations required to meet an error tolerance.

<u>example</u>: Suppose we need to find a root of
$$f(x) = x - 0.2\sin(x) - 0.5$$
with a final absolute error bound of $|x_i - x^*| \leq 5\times10^{-7}$

Since $f(0) = -0.5$ and $f(1) = +0.3317$ and $f(t)$ is continuous, we know that there is a root in $[0,1]$. How many iterations of the bisection method are required to meet the error tolerance?

<u>Sol$^{\underline{n}}$</u>: From what we derived last time, we need to find an n such that

$$e_n \leq \frac{1}{2^n}(1-0) \leq 5\times10^{-7}$$

$$\therefore \text{ want } 2^n \geq 0.2\times10^{-7}$$
$$n \geq \log(0.2\times10^{-7})/\log 2$$
$$\doteq 20.93$$

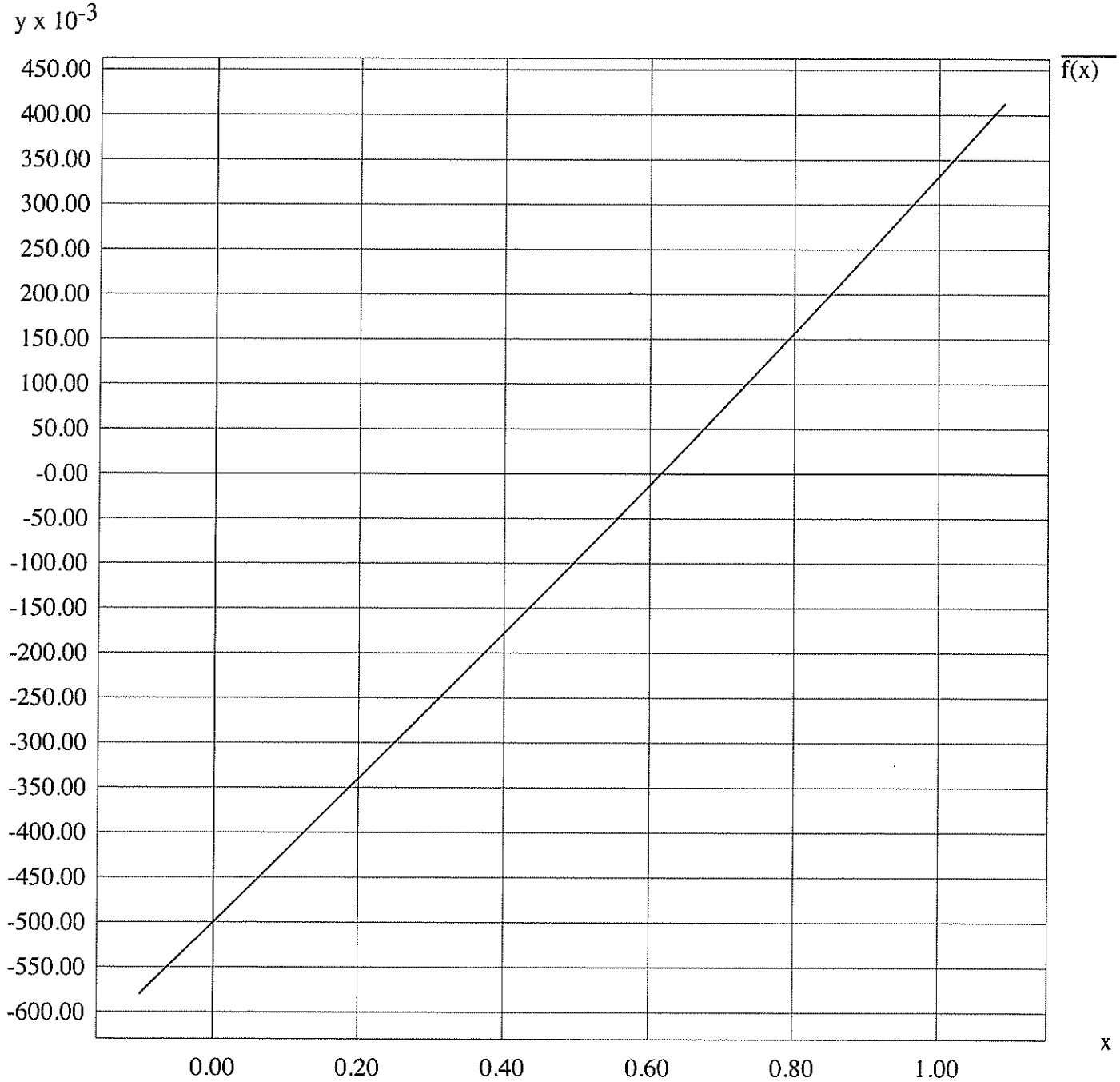
$\therefore$ we need 21 iterations of the bisection method.

Confirmed by results in handout. Note how $f(x_i)$ bounces around. This function ~ linear!
$\Rightarrow$ we should explore other methods!

Bisection applied to $f(x) = x - 0.2\sin(x) - 0.5$

step number	L	$f(L)$	R	$f(R)$	M	$f(M)$	max error
1	0.000000E+00	-	1.000000E+00	+	5.000000E-01	-9.588511E-02	5.0E-01
2	5.000000E-01	-	1.000000E+00	+	7.500000E-01	1.136722E-01	2.5E-01
3	5.000000E-01	-	7.500000E-01	+	6.250000E-01	7.980545E-03	1.2E-01
4	5.000000E-01	-	6.250000E-01	+	5.625000E-01	-4.416053E-02	6.2E-02
5	5.625000E-01	-	6.250000E-01	+	5.937500E-01	-1.814463E-02	3.1E-02
6	5.937500E-01	-	6.250000E-01	+	6.093750E-01	-5.096014E-03	1.6E-02
7	6.093750E-01	-	6.250000E-01	+	6.171875E-01	1.438734E-03	7.8E-03
8	6.093750E-01	-	6.171875E-01	+	6.132812E-01	-1.829518E-03	3.9E-03
9	6.132812E-01	-	6.171875E-01	+	6.152344E-01	-1.956125E-04	2.0E-03
10	6.152344E-01	-	6.171875E-01	+	6.162109E-01	6.215054E-04	9.8E-04
11	6.152344E-01	-	6.162109E-01	+	6.157227E-01	2.129326E-04	4.9E-04
12	6.152344E-01	-	6.157227E-01	+	6.154785E-01	8.656611E-06	2.4E-04
13	6.152344E-01	-	6.154785E-01	+	6.153564E-01	-9.347883E-05	1.2E-04
14	6.153564E-01	-	6.154785E-01	+	6.154175E-01	-4.241132E-05	6.1E-05
15	6.154175E-01	-	6.154785E-01	+	6.154480E-01	-1.687741E-05	3.1E-05
16	6.154480E-01	-	6.154785E-01	+	6.154633E-01	-4.110413E-06	1.5E-05
17	6.154633E-01	-	6.154785E-01	+	6.154709E-01	2.273095E-06	7.6E-06
18	6.154633E-01	-	6.154709E-01	+	6.154671E-01	-9.186596E-07	3.8E-06
19	6.154671E-01	-	6.154709E-01	+	6.154690E-01	6.772177E-07	1.9E-06
20	6.154671E-01	-	6.154690E-01	+	6.154680E-01	-1.207210E-07	9.5E-07
21	6.154680E-01	-	6.154690E-01	+	6.154685E-01	2.782483E-07	4.8E-07

f(x) = x - 0.2 sin(x) - 0.5



Fixed Point Iteration

where we solve an equivalent problem.

A problem that is equivalent to

find x^* s.t. $f(x^*) = 0$

is the problem

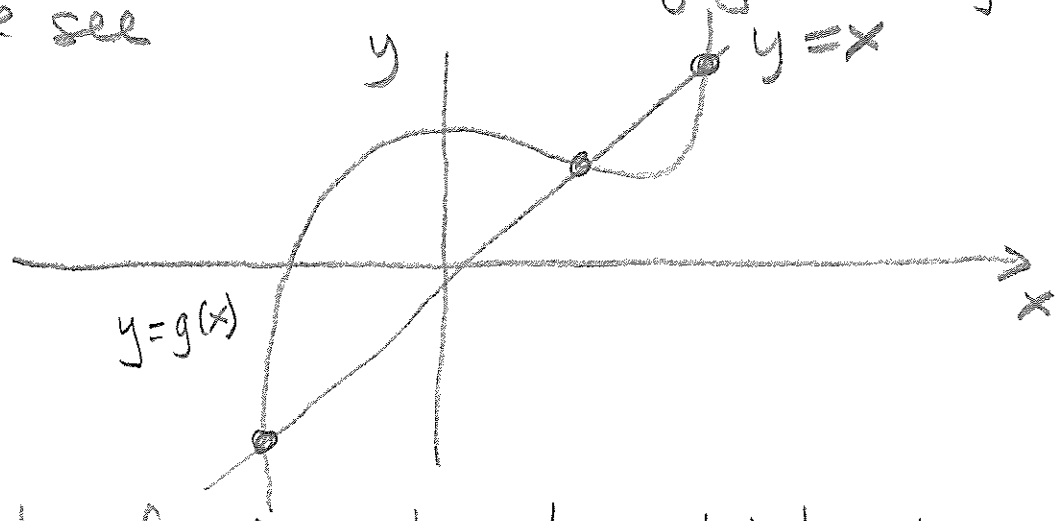
find p^* s.t. $g(p^*) = p^*$
where $g(x) = x - f(x)$

[since $g(p^*) = p^*$
gives $p^* - f(p^*) = p^*$
or $f(p^*) = 0$]

def$^{\underline{n}}$: A point p^* is called a fixed point of a function $g(x)$ if and only if $g(p^*) = p^*$.

We see that fixed points of $g(x)$ are roots of $f(x)$. Perhaps we can find roots of $f(x)$ by finding fixed points of $g(x)$.

Note: If we draw a sketch of $y = g(x)$ and $y = x$ we see



$g(x)$ has fixed points where it intersects the line $y = x$.

no intersection $\Rightarrow$ no fixed point

How can we find fixed points of $g(x)$?

One approach (called fixed point iteration or functional iteration) is to define the iteration sequence

$$x_0 \text{ given}$$
$$x_{i+1} = g(x_i) \quad \text{for } i = 0, 1, 2, \ldots$$

If this sequence converges, then it converges to a fixed point of $g(x)$.

Let's return to our previous problem of finding a root of

$$f(x) = x - 0.2\sin(x) - 0.5$$

(we found $x^* = 0.6514685$ using bisection in 21 steps)

How can we turn this into a fixed point problem?

Define $g(x) = 0.2\sin(x) + 0.5$

$\therefore f(x) = x - g(x)$

If we find a fixed point of $g(x)$, we will have found a root of $f(x)$.

Compute $x_0 = 0$
then $x_{i+1} = g(x_i)$

until $|x_{i+1} - x_i| < 5 \times 10^{-7}$

We get the following results $\longrightarrow$

i	x_i	change in x_i
0	0.000000E+00	
1	5.000000E-01	5.0E-01
2	5.958851E-01	9.6E-02
3	6.122483E-01	1.6E-02
4	6.149418E-01	2.7E-03
5	6.153822E-01	4.4E-04
6	6.154541E-01	7.2E-05
7	6.154659E-01	1.2E-05
8	6.154678E-01	1.9E-06
9	6.154681E-01	3.1E-07

decr.

Note that this converges in 9 steps (so much faster than bisection)

Furthermore, if we had started with

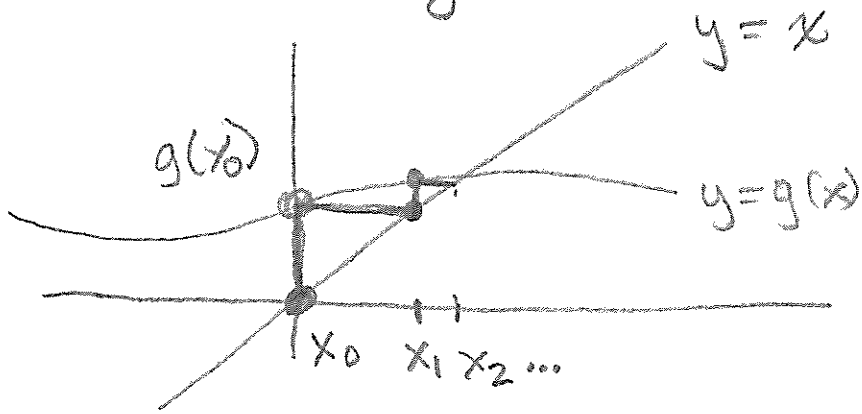
x_0	get convergence in
1	9 steps
-1	10
10	10
-100	7
1000	9
-1000	10

for this example

So the convergence rate does not depend much on the accuracy of the initial guess. (Compare with bisection, where # iteration depends on width of the initial interval]

We can visualize the convergence

- draw a graph of $y=x$ and $y=g(x)$
- start at $x_0=0$
- evaluate $g(x_0)$
- $x_1 = g(x_0)$
  draw line to $y=x$
- evaluate $g(x_1)$
- $x_2 = g(x_1)$



and so on. We move closer to the fixed point.

<u>example</u> Use fixed point iteration to find a root of $f(x) = x^3 - x - 1$

<u>first try</u> : $f(x) = (x^3 - 1) - x$

(isolate x) $= g(x) - x$

for $g(x) = x^3 - 1$

"backwards" form from before but this does not matter.

Try functional iteration on $g(x)$ from $x_0 = 1$.

$g(x^*) = x^*$
$\Rightarrow f(x^*) = 0$

We get:

i	x_i	change in x_i
0	1.000000E+00	
1	0.000000E+00	1.0E+00
2	-1.000000E+00	1.0E+00
3	-2.000000E+00	1.0E+00
4	-9.000000E+00	7.0E+00
5	-7.300000E+02	7.2E+02
6	-3.890170E+08	3.9E+08
7	-5.887159E+25	5.9E+25
8	-2.040409E+77	2.0E+77
9	-8.494771+231	8.5+231

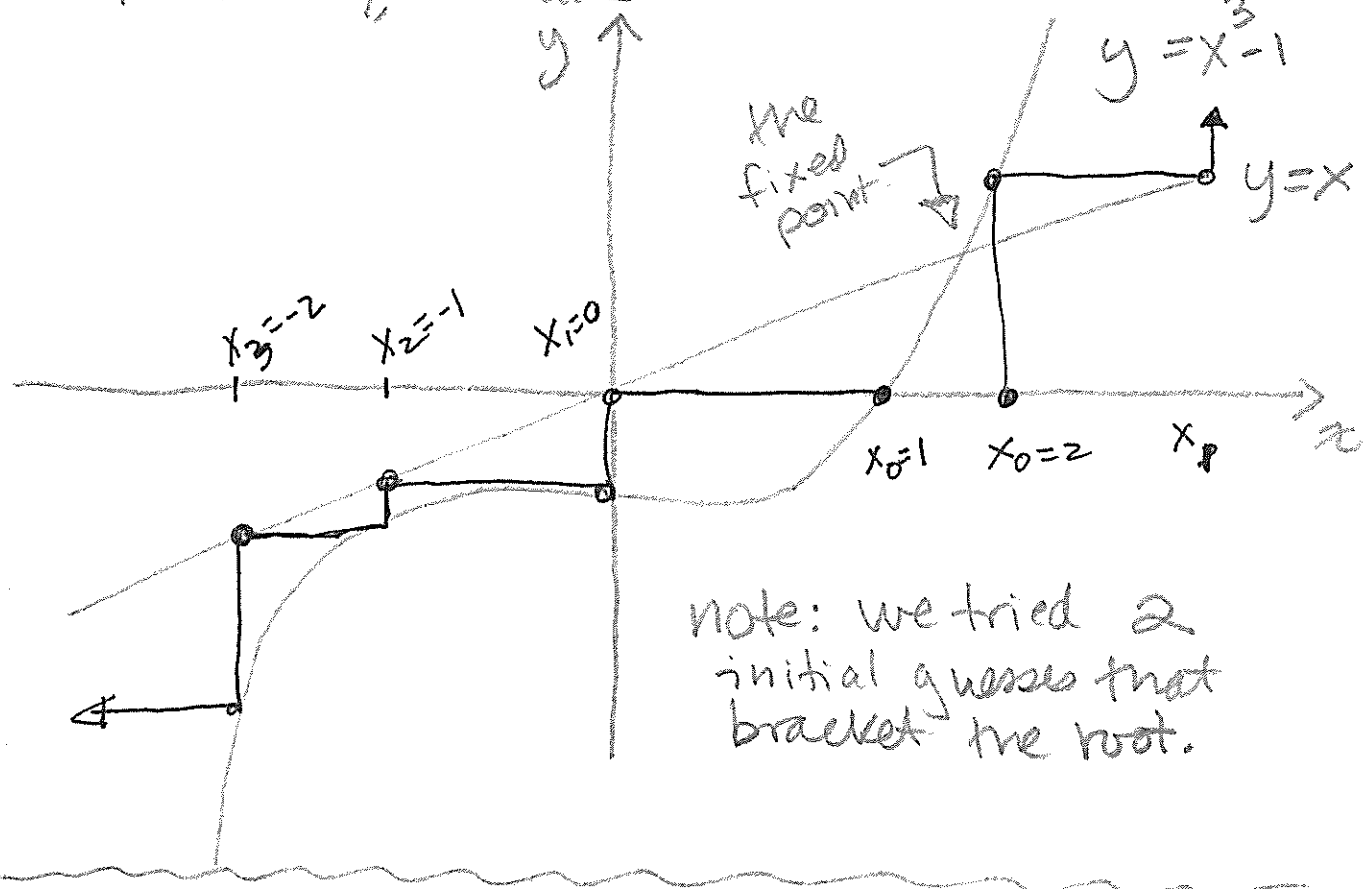
our iteration is blowing up!

Maybe just bad luck. Try again from $x_0 = 2$

i	x_i	change in x_i
0	2.000000E+00	
1	7.000000E+00	5.0E+00
2	3.420000E+02	3.4E+02
3	4.000169E+07	4.0E+07
4	6.400810E+22	6.4E+22
5	2.622435E+68	2.6E+68
6	1.803492+205	1.8+205

blowing up again! even faster!

Pictorally, we have



note: we tried 2 initial guesses that bracket the root.

A second try:

Alternatively we could try

$$f(x) = x^3 - (x+1)$$

which is $= 0$ if and only if

$$x^3 = x+1$$

or $$x = (x+1)^{1/3}.$$

Note: this $g(x) = (x+1)^{1/3}$ is not such that $g(x) = x - f(x)$. However fixed points of this $g(x)$ are roots of $f(x)$.

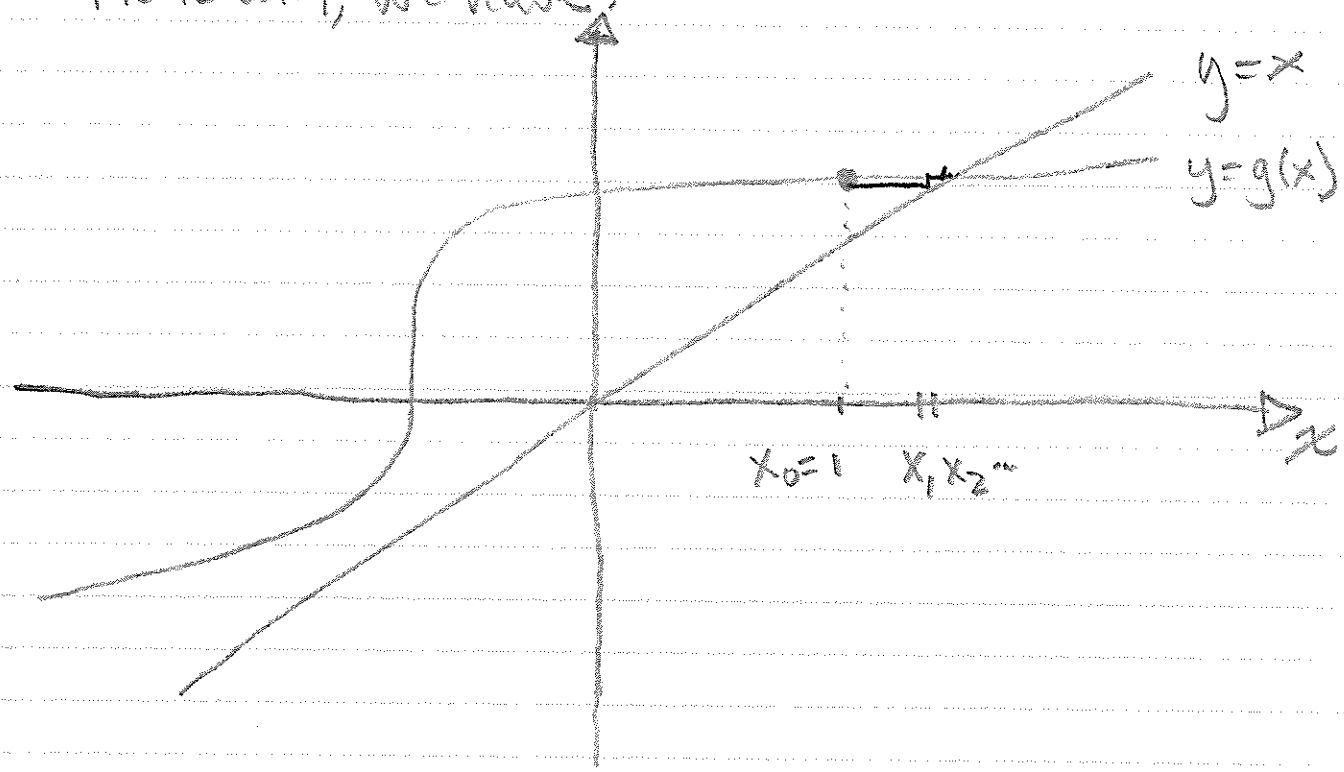
so take g(x) = (x+1)^(1/3)

Try functional iteration with $x_0 = 1$ gives: Convergence!

and $f(1.324718) \doteq 0$!

i	x_i	change in x_i
0	1.000000E+00	
1	1.259921E+00	2.6E-01
2	1.312294E+00	5.2E-02
3	1.322354E+00	1.0E-02
4	1.324269E+00	1.9E-03
5	1.324633E+00	3.6E-04
6	1.324702E+00	6.9E-05
7	1.324715E+00	1.3E-05
8	1.324717E+00	2.5E-06
9	1.324718E+00	4.7E-07

Pictorally, we have:



Summary:

- not all fixed point iterations solve the problem but some do.
- not all fixed points can be found by fixed point iteration

∴ we need to learn more about fixed points of functions and the fixed point iteration process.

Reasonable questions:

- when can we know function $g(x)$ has a fixed point?
- when do we know it is unique?
- when do we know that the fixed point iteration converges to a fixed point?
- when do we know that it won't converge?
- rate of convergence?